Class Modeling Project
ATMO5332 Fall 2026

You will be conducting a class project using the WRF-ARW V4 model that you used for HW#1.

General Guidelines

- Choose a scientific question that you want to answer with your modeling project. Remember, this is regional-scale NWP - make sure you have an appropriate topic.
- Start early! You may need to change your scope if something doesn't work, and queue waiting times can slow you down.
- Conducting the project in concert with your thesis research, if possible, is an excellent idea and encouraged, the only rule is that it can't be research you've already done.
- Your project must include quantitative analysis to support the answering of your research question.
- Minimum number of simulations is 6.
- GFS forecast data (and other data) is available through the AWS archive.

Key Date – October 1

Fill out the project plan (found in the "Project" module in Canvas) and turn it in to me by this date. You will be providing specific information on the following questions:

- What is your project scientific question(s)?
- What simulations are you going to perform?
- What quantitative analysis are you going to conduct?

I will provide feedback to ensure you are executing a sufficient depth of modeling and analysis.

Oral Presentation Guidelines

- Oral presentations should last about 12 minutes (please do not go over), and there will be a couple of minutes for questions.
- Prepare presentations as you would for a scientific conference, including an **Introduction** section stating and explaining your scientific question, a **Methodology** section (how are you answering your scientific question), a **Results** section (what are the answers to your scientific question**)**, and a **Conclusions** (summary) section.
- Your grade for the project will depend on designing and executing the appropriate modeling simulations (40 points) and quantitative analysis (40 points) to answer your research question. The final 20 points will depend on the clarity and quality of your presentation.

Presentation Schedule

November 19 – Sam

November 19 – Nelson

November 19 – Abid

November 19 – Rachel

November 19 - Min

November 24 – Luke

November 24 – Sydney

November 24 – Emily

November 24 – Lauren M.

November 24 – Shengshen

December 1 – Lauryn F.

December 1 – Rashid

December 1 – Logan

December 1 – Elliott

December 1 – Josiah